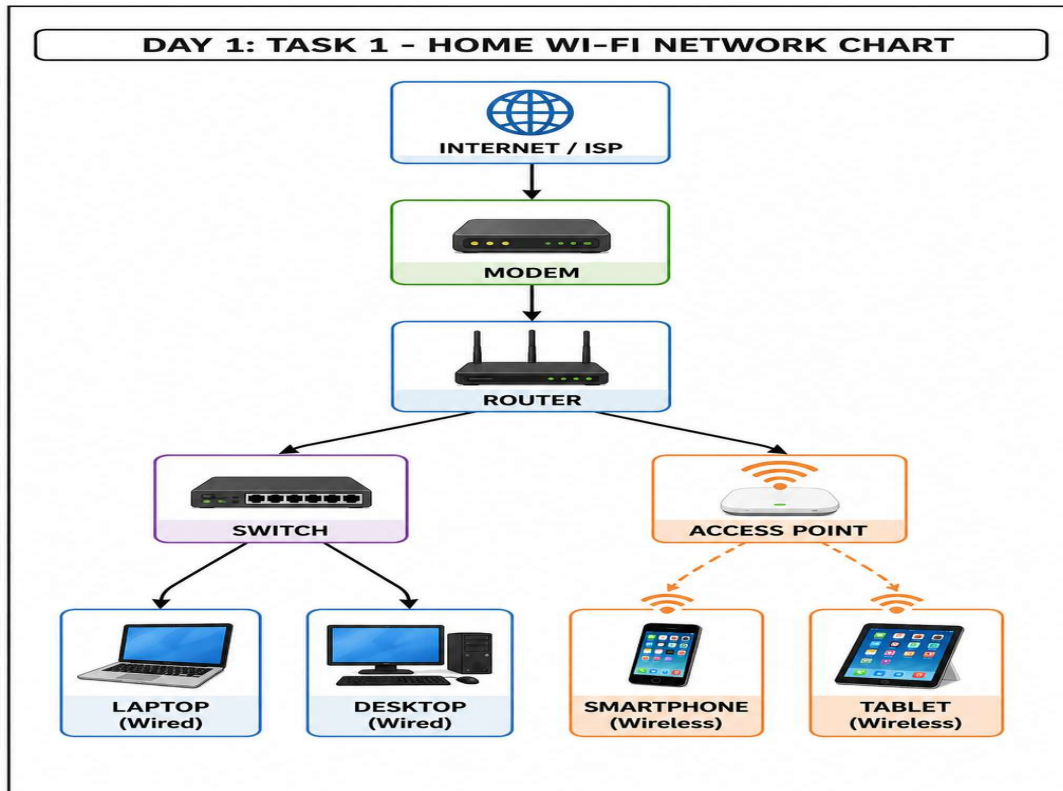


Task 1:

Home Wi-Fi Network Setup Diagram



2. Component Description & Signal Flow:

- **Internet Service Provider (ISP):** Delivers the external broadband internet connection into the home via Fiber/Coaxial cable.

- **Modem:** Converts incoming analog signals from the ISP into digital signals that home network hardware can process.
- **Router:** Connects the local home network (LAN) to the internet (WAN), assigns IP addresses via DHCP, and routes data packets.
- **Switch:** Expands the number of wired Ethernet ports available on the network to connect devices like laptops and desktop PCs using cables.
- **Access Point (AP):** Emits wireless Wi-Fi signals so wireless devices like smartphones and tablets can connect to the network.
- **End-User Devices (Laptop & Smartphone):** Laptops connect via Ethernet cable through the switch or wirelessly via the Access Point, while smartphones connect wirelessly to the Access Point.

